

MATHEMATICAL ANALYSIS 1
HOMEWORK 14

(1) For the following ODEs, answer these questions (explain your answer):

- (i) what is the order?
- (ii) is it linear?
- (iii) is it homogeneous?
- (iv) is it autonomous?

- (a) $y^{(5)} + (\sin x)y'' - \frac{3}{4}y = 0$
- (b) $y^{(5)} + (\sin x)y'' - \frac{3}{4}x = 0$
- (c) $y^{(5)} + (\sin x)y'' = \frac{3}{4}x$
- (d) $y^5 + (\sin x)y'' - \frac{3}{4}x = 0$
- (e) $y^{(5)} + (\sin y)y'' - \frac{3}{4}y = 0$

(2) **First-order ODEs.** Solve the following first-order separable ODEs:

- (a) $y' = x \ln(1 + x^2)$
- (c) $y' = \frac{y^2}{x \ln x} - \frac{1}{x \ln x}$
- (b) $y' = \frac{(x+2)y}{x(x+1)}$
- (d) $y' = \sqrt[3]{2y + 3} \tan^2 x$

(3) **First-order initial-value problems.** Solve the following initial-value problems:

- (a) For as long as the solution can be continued from $x = 0$, solve

$$\begin{cases} y' = \frac{1-e^{-y}}{2x+1}, \\ y(0) = 1. \end{cases}$$

- (b) In the interval $x \in (-2, 2)$, solve

$$\begin{cases} y' = \frac{3x}{x^2-4}|y|, \\ y(0) = -1. \end{cases}$$

- (c) For as long as the solution can be continued from $x = 1$, solve

$$\begin{cases} \frac{y''}{(1+y')^{3/2}} = \frac{8x^3}{(x^4+1)^2}, \\ y(1) = 0, \quad y'(1) = 3. \end{cases}$$

(hint: rename $z = y'$)

(4) **Second-order ODEs.** Solve the following second-order linear homogeneous ODEs:

- (a)

$$\begin{cases} y'' + 2y' + 5y = 0, \\ y(0) = 0, \quad y'(0) = 2. \end{cases}$$

- (b)

$$\begin{cases} y'' - 4y' + 4y = 0, \\ y(0) = 1, \quad y'(0) = 0. \end{cases}$$

- (c)

$$\begin{cases} y'' - 3y' + 2y = 0, \\ y(0) = 2, \quad y'(0) = 3. \end{cases}$$